

COL2010: Data Science

End-Semester Examination — Module 2: Data Cleaning and Preprocessing

SET B

Time: 3 Hours

Total Marks: 100

Instructions to candidates.

- This paper contains **15** questions. Attempt **all** questions.
- All numerical answers must retain at least four significant figures unless an exact closed form is requested.
- All claims of the form “prove”, “show”, or “derive” require a fully rigorous argument; a correct final formula with no justification earns no credit.
- State any assumption you invoke that is not already given in the question.
- Calculators are permitted; symbolic computer algebra devices are not.

Q1. For two non-zero vectors $x, y \in \mathbb{R}^d$ define the cosine distance $d_{\cos}(x, y) = 1 - \frac{x^\top y}{\|x\|_2 \|y\|_2}$.

- (a) Verify that $d_{\cos}$ is non-negative and symmetric, and identify precisely the set of pairs (x, y) for which $d_{\cos}(x, y) = 0$. Explain why this shows $d_{\cos}$ is at best a *pseudometric* on $\mathbb{R}^d \setminus \{0\}$, not a true metric.
- (b) Consider the three vectors $a = (1, 0)$, $b = (0, 1)$, $c = (1, 1)$. Compute $d_{\cos}(a, b)$, $d_{\cos}(a, c)$, and $d_{\cos}(b, c)$ exactly, and show explicitly that the triangle inequality $d_{\cos}(a, b) \leq d_{\cos}(a, c) + d_{\cos}(c, b)$ **fails**.
- (c) Define instead the angular distance $\theta(x, y) = \arccos\left(\frac{x^\top y}{\|x\|_2 \|y\|_2}\right) \in [0, \pi]$. Using the fact that θ measures the geodesic distance between the corresponding points on the unit sphere, argue why θ *does* satisfy the triangle inequality (a full spherical-geometry proof is not required, but your argument must be more than an assertion), and verify numerically, using your three vectors a, b, c , that $\theta(a, b) \leq \theta(a, c) + \theta(c, b)$.

[8 marks]

Q2. A laboratory records serum creatinine (mg/dL) for $n = 9$ patients in one ward:

0.8, 0.9, 0.9, 1.0, 1.0, 1.1, 1.1, 1.2, 4.5.

Compute Q_1 , the median, Q_3 , and IQR using the median-of-halves convention. Hence compute the lower and upper Tukey fences, and state which observation(s) are flagged.
[4 marks]

Q3. As in Q3 of the general syllabus, consider a column in which a fraction $\varepsilon \in (0, 1)$ of n readings are coordinated instrument-fault values, all equal to M , while the remaining readings are confined to a bounded, effectively negligible range around 0 (relative to $M \rightarrow \infty$).

- (a) Derive $\mu(M)$ and $\sigma(M)$ in this limit, and show that the ordinary z -score of a faulty reading, $z(M) = (M - \mu(M))/\sigma(M)$, tends to the finite limit $\sqrt{(1 - \varepsilon)/\varepsilon}$ as $M \rightarrow \infty$.
- (b) Solve for the critical fault fraction ε^* at which this limit equals exactly 3, and interpret clinically: state, in one sentence, what your result implies about a batch where more than ε^* of a ward's instrument readings are corrupted identically.

- (c) Prove that the modified z -score based on median and MAD instead diverges to $+\infty$ as $M \rightarrow \infty$, for any fixed $\varepsilon < 0.5$, and explain precisely why the *breakdown point* of (mean, std) should not naively be quoted as “any single outlier defeats it” when discussing a *fixed-fraction* contamination model such as this one — distinguish the single-point notion from the fixed-fraction asymptotic notion.

[9 marks]

Q4. Two patients are described by (Age in years, Total Cholesterol in mg/L): $P = (45, 2800)$, $Q = (62, 3100)$.

- Compute the raw Euclidean distance and identify the dominating coordinate.
- Given ward statistics $\mu_{\text{age}} = 50, \sigma_{\text{age}} = 12$ and $\mu_{\text{chol}} = 2900, \sigma_{\text{chol}} = 350$, standardize both coordinates of P and Q and recompute the Euclidean distance.
- A k -NN-based cohort-matching tool is run *before* scaling. Explain, quoting your two numerical distances, why cholesterol (in these units) would completely dominate every neighbour search regardless of clinical relevance.

[6 marks]

Q5. Two patients’ presenting symptoms are encoded over the tag set {Fever, Cough, Fatigue, Nausea, Rash} as binary vectors $x = (1, 1, 0, 0, 1)$ and $y = (1, 0, 0, 1, 1)$.

- Compute the Jaccard similarity and cosine similarity of x and y .
- A third patient’s record $z = (0, 0, 0, 0, 0)$ has no symptoms recorded (a data-entry placeholder, not a clinical claim of perfect health). State $\cos(z, y)$ and $\cos(z, z)$ under the course convention, and explain why treating z as a genuine “no symptoms” observation rather than a missingness placeholder could silently corrupt a similarity-based triage tool.

[4 marks]

Q6. Let R be the missingness indicator for a clinical variable $Y = (Y_{\text{obs}}, Y_{\text{mis}})$, with mechanism $P(R \mid Y_{\text{obs}}, Y_{\text{mis}}; \phi)$.

- State Rubin’s formal definitions of MCAR, MAR, and MNAR.
- Prove the ignorability result: under MAR and distinctness of parameters, the observed-data likelihood factors so that likelihood/Bayesian inference need not model ϕ explicitly.
- A ward observes that patients experiencing the most severe pain are *less* likely to complete a self-reported pain-scale form (severe pain impairs form completion), so $P(R = 0 \mid \text{PainScore})$ increases in the true (unrecorded) PainScore itself. Classify this mechanism formally and explain concretely, via your proof in (b)-(c), why no imputation using only observed columns can remove the resulting bias.
- Suppose the hospital additionally records a nurse-observed distress rating for every patient (always present, never missing). Explain, in general terms, how conditioning on this additional observed variable could in principle convert the mechanism in (c) into one that is MAR *relative to the enlarged set of observed variables*, and why this shows that the MCAR/MAR/MNAR classification is relative to which variables one has actually observed, not an intrinsic property of the missing variable alone.

[8 marks]

- Q7.** An analyst applies iterative predictive imputation to five patient records over columns Age, BMI, and Systolic Blood Pressure (SBP):

Row	Age	BMI	SBP
1	40	24	118
2	55	29	?
3	33	22	110
4	?	31	142
5	61	27	150

Missing cells are initialized with the column mean over observed values. The cyclic order is **SBP** $\rightarrow$ **Age**, using the fitted relations

$$\widehat{\text{SBP}} = 0.8 \cdot \text{Age} + 1.5 \cdot \text{BMI} + 70, \quad \widehat{\text{Age}} = 0.3 \cdot \text{SBP} - 2 \cdot \text{BMI} + 40.$$

- State the two initial fill values.
- Perform one full iteration (update Row 2's SBP, then Row 4's Age), showing every substituted number.
- Explain, referencing the specific predictor columns used for each target here, why a second iteration would leave both values unchanged, and give a general condition on predictor overlap under which convergence would instead require multiple passes.

[6 marks]

- Q8.** A hospital record-linkage system computes pairwise similarities for three patient records A, B, C across two source systems: $s(A, B) = 0.80$, $s(B, C) = 0.78$, $s(A, C) = 0.15$, with lower threshold 0.35 and upper threshold 0.70.

- Classify each pair (nonmatch / review / match).
- State the connected-components clustering of $\{A, B, C\}$ using only “match” edges, and explain the resulting transitivity problem.
- Using the objective $\text{Obj}(\mathcal{C}) = \sum_{i,j \text{ same cluster}} (s(i, j) - \theta)$ with $\theta = 0.5$, compute Obj for the connected-components clustering $\{A, B, C\}$ and for the alternative $\{A, B\}, \{C\}$, and conclude which is objectively better.
- Generalize to a path of n patient records where every adjacent pair scores s_0 (just above upper threshold) and every non-adjacent pair scores a floor f (below lower threshold). Derive Obj exactly as a function of n, s_0, f, θ for (i) the single connected-component clustering and (ii) the clustering that keeps only adjacent pairs together, and prove the gap between them is $\Theta(n^2)$. State the general moral about transitive closure over non-transitive pairwise decisions in an identity-resolution system spanning many hospital feeds.

[10 marks]

- Q9.** Hospital bill amounts (in Rs. '000) for ten patients are strongly right-skewed: 8, 12, 15, 22, 30, 55, 120, 300, 500, 1000. Compute $\log_{10}$ of each value to two decimal places, and state the raw ratio $\max / \min$ against the transformed range $\log_{10}(\max) - \log_{10}(\min)$, quantifying the compression achieved.

[4 marks]

- Q10.** A hospital analytics team fits a median imputer for BMI and a one-hot encoder for a categorical *Diagnosis Code* column on the entire dataset (training and test rows combined) before splitting.

- (a) With $n_{\text{tr}}, n_{\text{te}}$ the training/test sizes and $\mu_{\text{tr}}, \mu_{\text{te}}$ the true population BMI means restricted to each subpopulation (which may genuinely differ, e.g. if test data come from a later time period with a different patient mix), derive the exact bias $\mathbb{E}[\hat{\mu}_{\text{full}}] - \mu_{\text{tr}}$ of the leaked estimator relative to the correct train-only estimator, and state the condition under which it vanishes.
- (b) Explain formally why this bias implies any downstream test-set performance metric is an optimistically biased estimate of true generalization performance.
- (c) A test-time patient carries a rare Diagnosis Code that never appeared among training rows. State precisely how a correctly leak-free one-hot encoder (fitted on training categories only) must represent this row, and explain why fitting the encoder on the union of all codes seen across the whole dataset is still leakage, even though it involves no numerical statistic at all.

[8 marks]

Q11. Two candidate duplicate patient records are compared on four fields: name similarity = 0.9 (Jaro-Winkler), date-of-birth agreement = 1, phone agreement = NaN (one phone missing), address similarity = 0.6. Field weights are name 0.35, DOB 0.30, phone 0.20, address 0.15.

- (a) Compute the weighted-mean score using only available fields, with weights renormalized.
- (b) Classify against thresholds lower = 0.5, upper = 0.8.
- (c) Explain why substituting 0 for the missing phone comparison, instead of excluding it and renormalizing, would bias the decision, and in which direction.

[6 marks]

Q12. A wearable-device metric (minutes of continuous monitoring per session) for $n = 7$ patients reads: 70, 72, 71, 73, 70, 72, 140.

- (a) Compute the mean, standard deviation, and ordinary z -score of the value 140.
- (b) Compute the median, MAD, and modified z -score of the value 140.
- (c) Using $|z| \geq 3$ and $|z_{\text{mod}}| > 3.5$, state which method(s) flag 140, and connect the disagreement explicitly to your general result in Q3.

[4 marks]

Q13. For $x_1, x_2 \in \mathbb{R}$ and $0 < p < 1$, define $\|(x_1, x_2)\|_p = (|x_1|^p + |x_2|^p)^{1/p}$.

- (a) Verify that the points $u = (1, 0)$ and $v = (0, 1)$ both satisfy $\|u\|_{1/2} = \|v\|_{1/2} = 1$.
- (b) Compute $\|\frac{1}{2}u + \frac{1}{2}v\|_{1/2}$ exactly, and show it exceeds 1.
- (c) Explain precisely why part (b) proves that the “unit ball” $\{x : \|x\|_{1/2} \leq 1\}$ is **not convex**, and hence that $\|\cdot\|_{1/2}$ cannot be a norm (recall that every norm’s unit ball must be convex, a consequence of the triangle inequality applied to $\|\lambda u + (1 - \lambda)v\| \leq \lambda\|u\| + (1 - \lambda)\|v\|$).
- (d) Contrast this with the case $p \geq 1$: state (without a full proof) which standard inequality guarantees convexity of the ℓ_p unit ball for $p \geq 1$, and hence why fractional-order “Minkowski distances” with $p < 1$ are never used as genuine distance metrics in the outlier-detection pipelines discussed in this course.

[8 marks]

Q14. Two blocking passes generate duplicate-candidate pairs among $n = 5000$ standardized patient records. Pass 1 (same admitting ward and birth year) yields blocks of sizes 60, 90, 35. Pass 2 (same last-four contact-number digits) yields blocks of sizes 10, 14, 8, 22. The two candidate sets overlap in exactly 50 pairs.

- Compute the candidate pairs generated by each pass and the size of their union.
- Compute the reduction ratio RR for the union relative to $\binom{n}{2}$.
- Ground truth contains 400 true duplicate pairs; Pass 1 alone captures 350, Pass 2 alone captures 220, and both jointly capture 170. Verify via inclusion–exclusion that the union achieves pair completeness $PC = 1.0$, i.e. perfect recall, and explain in words why this can happen even though neither pass alone is complete.
- Prove the general claim $1 - (1 - p_1)(1 - p_2) \geq \max(p_1, p_2)$ for capture probabilities $p_1, p_2 \in [0, 1]$ of two independent blocking passes, and relate it to the phenomenon you observed in (c).

[6 marks]

Q15. A hospital data-quality officer gives you eight patient records:

ID	Age	Height (cm)	Weight (kg)	SBP (mmHg)	Name (standardized)
1	34	170	68	118	ravi kumar
2	29	165	60	122	ravi kumar
3	45	168	65	?	meena iyer
4	51	172	70	150	suresh babu
5	38	180	42	116	farah khan
6	60	166	63	?	j. verma
7	41	171	66	121	meena iyer
8	33	169	64	119	rita das

Among the six rows with observed SBP, mean Age is 37.67; among the two with missing SBP, mean Age is 52.5. From the other seven rows (excluding Row 5), $\mu = (\mu_H, \mu_W) = (169.4, 65.4)$ and $\Sigma = \begin{bmatrix} 16.8 & 9.5 \\ 9.5 & 7.2 \end{bmatrix}$. Rows 3 and 7 are suspected duplicates, with comparison vector: name similarity 0.95, phone NaN, ward agreement 1, DOB agreement 0, using the weights/thresholds of Q11.

- State, with the standard caveat about MNAR being unprovable from observed data alone, what the age-conditioned evidence does and does not establish about the missingness mechanism of SBP.
- Compute, by hand, the Mahalanobis distance of Row 5's $(H, W) = (180, 42)$ from μ , explicitly computing Σ^{-1} , and comment on whether Row 5 is a multivariate outlier despite each coordinate looking unremarkable alone.
- Decide, via the Q11 machinery, whether Rows 3 and 7 should be merged, and construct the merged record's SBP field using the course's numeric golden-record rule.
- Argue formally whether deduplication should precede or follow imputation here, prove that your ordering strictly dominates the reverse for at least one field, and state the audit-trail invariant that must survive regardless of ordering.

[9 marks]